the temperature of the hot reservoir from 300°C to 350°C by keeping the cold reservoir temperature constant. Which is the suitable method?

Answer: Initial efficiency = 34.9%

Efficiency in method (a) =43.6%

Efficiency in method (b) =40.1%

Method (a) is more efficient.

14. A Carnot engine whose efficiency is 45% takes heat from a source maintained at a temperature of 327°C. To have an engine of efficiency 60% what must be the intake temperature for the same exhaust (sink) temperature?

Answer: 552°C

15. An ideal refrigerator keeps its content at 0°C while the room temperature is 27°C. Calculate its coefficient of performance.

Answer: β=10.11

BOOKS FOR REFERENCE

1. Serway and Jewett, Physics for scientist and Engineers with modern physics, Brook/Coole publishers, Eighth edition
2. Paul Tipler and Gene Mosca, Physics for scientist and engineers with modern physics, Sixth edition, W.H.Freeman and Company
3. James Walker, Physics, Addison-Wesley publishers, 4th Edition
4. Douglas C Giancoli, Physics for scientist & Engineers with modern physics, Pearson Prentice Hall, 4th edition.
5. H.C.Verma, Concepts of physics volume 1 and Volume 2, Bharati Bhawan Publishers
6. Tarasov and Tarasova, Question and problems in school physics, Mir Publishers

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